Mini Project 1.1: results

1. Isolated execution time (step 2, 300 calls, varying inputs)

The observed maximum is a measured, not proven, WCET estimate: no test campaign guarantees the longest path and worst hardware state were exercised (Section 1.7). A static analysis gives a sound bound instead.

Mean (ms)	Std (ms)	Min (ms)	Max (ms)	95th pct (ms)
15.96	5.23	4.07	40.33	24.89

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2. Periodic runs (T = 50 ms, D = 40 ms, 150 jobs)

Release jitter = actual - ideal release. Completion jitter = deviation of (completion - ideal release) from its mean. Utilities are per-job means.

Metric	Unloaded	Background thread
Mean response R (ms)	16.54	26.00
Max response R (ms)	49.41	79.19
Release jitter std (ms)	0.04	3.00
Release jitter max |.| (ms)	0.55	34.35
Completion jitter std (ms)	6.14	11.51
Completion jitter max |.| (ms)	32.86	52.70
Deadline-hit fraction	98.7%	90.0%
Mean hard utility (penalty -10)	0.85	-0.10
Mean firm utility	0.99	0.90
Mean soft utility (tau = D/4)	0.99	0.95

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3. Lab Task 2: adversarial versus ordinary inputs

Adversarial inputs: maximum work size with the deep code path (150 calls).

Input set	Mean (ms)	95th pct (ms)	Max (ms)
Ordinary (random)	15.96	24.89	40.33
Adversarial	46.40	60.00	71.11

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4. Lab Task 3: soft-deadline decay rate (loaded run)

Decay setting	tau (ms)	Mean soft utility
fast decay (0.2 x tau)	2.0	0.913
baseline tau	10.0	0.950
slow decay (5 x tau)	50.0	0.984

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5. Lab Task 4: varying the period (D = 0.8 T, 100 jobs each)

T (ms)	D (ms)	Rel. jitter std (ms)	Rel. jitter max (ms)	Std (% of T)	Max (% of T)	Hit fraction
30	24	0.61	4.74	2.03	15.80	84.0%
50	40	0.03	0.31	0.06	0.61	100.0%
100	80	0.29	3.04	0.29	3.04	100.0%
200	160	0.03	0.29	0.02	0.14	100.0%